GenAI Literacy for CS1 Students

AI Adoption, Compliance and Societal Change

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Editorial

This is a working document which will be completed in September.

1 Introduction

The rise of Generative Artificial Intelligence (GenAI) has seen rapid global adoption, with the technology increasingly embedded across digital infrastructure. This is especially prevalent in computer science and programming: code editors now ship with built-in AI pair-programmers (GitHub Copilot, Cursor), office applications embed AI assistants (Microsoft Copilot, Google Gemini in Workspace), and search engines increasingly answer queries with AI-generated summaries rather than links to sources. For students entering computer science today, GenAI is not an optional add-on but an encompassing part of the tools they will use from their very first line of code.

While GenAI is touted by many as a step forward, a growing worry spreads through industry and academia about its effect on our own capabilities. One such worry is *deskilling*: the possibility that reliance on an AI tool causes the underlying human skill it assists with to atrophy. Studies of experts using AI tools in medicine [3] and software development [12] suggest that even skilled professionals can get worse at tasks their job requires as they grow more dependent on AI assistance. A second, related worry concerns the learning of such skills in the first place: studies of AI tool use in learning contexts, such as essay writing [8] and rudimentary research tasks [13], suggest these tools promote a shallower approach to learning, with learners often retaining less knowledge than peers using more traditional means, such as web search or no tool at all.

This is a particular concern for computer science education. GenAI tools have long been able to write programs expected of first-semester students in *Introductory Programming* (CS1); as early as 2022, OpenAI's Codex model solved the large majority of typical CS1 exam and exercise questions, often outperforming the average student [7]. Four years on, these tools have improved substantially and are now freely available to anyone with an internet connection. This ability to generate all of a student's code, paired with the risks of deskilling and shallow learning outlined above, raises significant concerns about non-learning in CS1 for any student unable to use these tools appropriately and safely.

1.1 Scope and Aim

This paper investigates how to address this situation and equip CS1 students with the information needed to use GenAI without compromising their learning outcomes, asking:

> **RQ**: What information about GenAI should be provided to CS1 students for them to use such tools without compromising learning in the course?

For this we turn to the concept of *AI Literacy*. Literacy is traditionally understood as the ability to identify, interpret, create, and communicate using written language [5], a concept since extended to describe comparable skill sets in other domains. *AI literacy* is one of the most recent such extensions, applying this same idea of enabling expression, communication, and access to knowledge to the domain of artificial intelligence [9].

The aim of this paper has been to develop a lecture delivered in the very first week of a CS1 course at the University of Bergen, fall 2026. I start by outlining the details of AI literacy, before providing content for a lecture of information for AI literacy competencies. I end with a reflection on how the lecture went and how it was perceived by the students.

2 Background

In their literature review, Long and Magerko [9] developed a definition for AI literacy:

> *a set of competencies that enables individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool online, at home, and in the workplace.*

This was paired with a framework of various competencies operationalizing their definition, covering five overarching themes/questions.

2.1 What is AI?

The first theme concerns the ability to recognize AI and situate it conceptually. It includes competencies such as distinguishing which technological artifacts do and do not use AI, and critically discussing what makes an entity "intelligent" at all—including how human, animal, and machine intelligence differ [9]. This theme also covers the interdisciplinary origins of AI as a field, and the distinction between *narrow* AI (systems designed to perform a single, specific task) and *general* AI (a hypothetical system capable of human-like reasoning across arbitrary domains). Much of the public discourse around "AI" conflates narrow, task-specific tools like code assistants with the far more speculative notion of general intelligence, and disentangling the two is a prerequisite for reasoning clearly about what these tools can and cannot be trusted to do [9].

2.2 What Can AI Do?

The second theme concerns AI's capabilities and limitations, i.e. the ability to identify what AI is currently good and bad at. Applied to GenAI in CS1: these tools are demonstrably capable of solving the large majority of typical introductory programming exercises [7], but they are equally capable of producing code that is subtly incorrect, inefficient, insecure, or stylistically poor while still appearing fluent and confident. A student who has not developed this competency has no basis for knowing when to trust a generated solution and when to interrogate it—which could lead to uncritical copy-pasting of GenAI output.

2.3 How Does AI Work?

The third theme is the most technically dense of the five: how AI systems represent information, make decisions, and learn from data, as well as the human role involved in building and training such systems [9]. For GenAI specifically, this means understanding that today's tools are built on deep neural networks—most commonly the transformer architecture—trained on vast quantities of text to predict the statistically most likely next word or code token, rather than to reason symbolically about correctness. Grasping this is what allows a student to understand *why* a model can generate plausible-looking code that nonetheless fails: the system has no explicit model of program semantics, only a learned statistical pattern over what code *typically* looks like given similar prior context.

2.4 How Should AI Be Used?

The fourth theme concerns the ethical and normative dimensions of AI use, encompassing considerations such as privacy, bias, transparency, and accountability [9]. While such aspects are important, our focus lies more in how to *learn with such tools*: what constitutes appropriate use of GenAI when the explicit goal of the course is for students to develop their own programming competency, where the line falls between using a tool to learn and using it to bypass learning entirely.

2.5 How Do People Perceive AI?

The fifth theme concerns how humans emotionally and socially relate to AI systems, including the tendency toward anthropomorphism and an understanding that AI agents are ultimately programmable systems rather than autonomous minds [9]. As conversational GenAI tools are specifically designed to communicate in a natural, confident, human-like register, it can lead novices (such as first-semester students) to over-attribute understanding or authority to their output. A student who perceives a chatbot's fluent explanation as equivalent to an instructor's verified expertise is less likely to question it—so counteracting this perception, and re-grounding these tools as programmed systems with no ground truth of their own, is itself a literacy goal.

3 Lecture Content

I now present the content that was delivered in the first lecture of the *Introductory Programming* course, fall 2026, at the Department of Information Science and Media Studies, University of Bergen. While this material is founded on Long and Magerko's AI literacy framework [9], the lecture is limited to 50 minutes and hence the content reflects a more narrow focus, instead of every competency outlined.

Since the medium of a paper such as this differs considerably from that of a lecture, the sections that follow are not written as a speaking script, but closer to a scientific paper. I therefore omit the showmanship required to hold students' attention in the room, and instead provide additional detail elaborating on the talking points themselves. The idea is that someone could read this and develop a lecture of their own.

3.1 What is AI?

A general definition of artificial intelligence, as given by Wikipedia, is [1]:

> *Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.*

Such a definition is quite broad and puts into question to what degree such technologies embody intelligence. Let us look at some examples.

Calculators

The term "computer" existed long before mechanical machines did our computations for us. It was rather reserved for humans who did calculations [**?**]. A famous example of this were the women who helped land us on the moon. Throughout the 1960s, a group of women at NASA's Langley Research Center worked as human "computers," performing by hand the trajectory calculations that NASA's electronic machines were not yet trusted to get right [**?**].

The job of "computer" is of course not in existence today, due to us developing calculators and programs that can conduct such calculations with more precision and speed than a human ever could. While this technology performs a task typically associated with human intelligence, few would claim a calculator is intelligent.

Chess

IBM's Deep Blue—a chess bot—offers a more contested case. In a 1996 six-game match, world chess champion Garry Kasparov defeated Deep Blue, winning three games to the machine's one, with two draws. Following an upgrade, Deep Blue met Kasparov again in a 1997 rematch and reversed the result, winning two games to Kasparov's one, with three draws—becoming the first computer to defeat a reigning world champion in a full match under tournament conditions, a result since regarded as a milestone in the history of artificial intelligence [2].

On the surface this seems to encapsulate human intelligence much more than a calculator: while I myself am able to do arithmetic and more advanced mathematics, I am far from able to beat a chess grandmaster. Yet a computer scientist would likely disagree that Deep Blue is intelligent in a human sense. The engine uses an algorithm called *minimax*: it builds out a tree of possible future moves several turns deep, scores each resulting board position with a heuristic evaluation function, and then picks the move that gives it the best guaranteed outcome, assuming its opponent always responds with their strongest possible counter-move [11]. There is no learning, no generalization, and no understanding of the game in a human sense here—only exhaustive, brute-force calculation guided by rules that human engineers wrote in advance.

GenAI

These days the term AI is used primarily to refer to GenAI tools such as Claude, ChatGPT, and Gemini—computational techniques capable of generating seemingly new, meaningful content such as text, images, or code from training data [6]. These tools have an underlying technology that can seem more complex than the previous two examples: they use *machine learning*. For us to understand how such tools are able to produce human-like text, we need to explore this underlying technique.

3.2 How Does AI Work?

Editorial:

> *Since this is the most complex part, the students were given homework before the lecture: watch 3Blue1Brown's videos on neural networks and large language models. These give detailed, but introductory information about this technology. The lecture then proceeds to talk about these techniques more generally.*

To understand how tools like Claude, ChatGPT, and Gemini can produce fluent, human-like text, we need to look at *machine learning* (ML): a branch of AI in which a system improves at a task not by following rules a human explicitly programmed—as Deep Blue's engineers did—but by identifying statistical patterns in large amounts of data. Rather than being told *how* to solve a problem, a machine learning model is shown many examples and learns a function that approximates the relationship between them.

Neural Networks

Most modern ML systems, including today's GenAI tools, are built from *artificial neural networks*: layers of simple mathematical units ("neurons") connected to one another, loosely inspired by neurons in the brain. Each connection has a numerical *weight*, and during training these weights are adjusted, incrementally and automatically, so that the network's output gets closer to some desired result. A network with many such layers is called a *deep* neural network, which is where the term *deep learning* comes from. Nobody hand-writes the rules that end up encoded in these weights; they emerge purely from repeated exposure to data.

From Neural Networks to Language

Large Language Models (LLMs)—the technology underlying GenAI chatbots—are deep neural networks built specifically to process and generate text. Since 2017, almost all state-of-the-art LLMs have used an architecture called the *transformer*, which introduced a mechanism called *self-attention*: for every word (or word-fragment, called a *token*) in a piece of text, the model computes how relevant every other token is to it, allowing it to capture relationships between words regardless of how far apart they appear in a sentence [15]. This is what allows an LLM to, for example, correctly resolve which noun a pronoun refers to several sentences earlier.

An LLM is trained by being shown enormous quantities of text scraped from the internet, books, and code repositories, with one simple task: given a sequence of tokens, predict the next one. It is never told any grammar rules, facts, or programming syntax directly; instead, by being trained to correctly guess the next token across billions of examples, the network's weights come to encode statistical regularities of language and code well enough to generate convincing, novel text one token at a time, each token chosen based on all tokens generated so far [15].

From text predictor to assistant

A model trained only to predict the next token is not yet a helpful assistant—left alone, it simply continues whatever text pattern it was given. To turn such a model into something like Claude or ChatGPT, developers apply an additional training stage in which human reviewers rank different model responses to the same prompt, and the model is further adjusted to produce the kinds of responses humans preferred—a technique known as *reinforcement learning from human feedback* [10]. This step is what makes the model behave like a helpful conversational partner rather than a raw autocomplete engine, though the underlying mechanism producing each word is still, fundamentally, a statistical prediction rather than an act of reasoning about truth or correctness.

Does ChatGPT Equate to Human Intelligence?

Whether a system that can fluently continue text should be considered intelligent in a human sense is exactly the question linguist Noam Chomsky, together with Ian Roberts and Jeffrey Watumull, take up in their widely discussed 2023 essay "The False Promise of ChatGPT" [4]. Their central argument rests on a distinction between three things a system can do when confronted with a physical event: *describe* it, *predict* it, and *explain* it.

Chomsky illustrates this with the example of a dropped apple. Observing that the apple falls is a description; stating in advance that it will fall once released is a prediction. Both are useful, and both can be correct, but neither is yet an explanation. A genuine explanation, Chomsky argues, requires a further step: a counterfactual claim that *any* such object would behave the same way, tied to a causal mechanism such as gravity or the curvature of space-time. Only this last move—reasoning from a general causal principle rather than from an observed instance—constitutes what Chomsky calls thinking. Tellingly, he notes that a causal explanation can, in principle, be wrong, since a claim only counts as an explanation if it is falsifiable [4].

Applied to GenAI, the argument follows directly from how LLMs are trained, as described above: an LLM learns to predict the next token by finding statistical regularities across enormous quantities of text, not by constructing an internal causal model of the world. It can therefore describe that apples fall, and predict that a given apple will fall, because both are patterns abundantly present in its training data. But Chomsky's claim is that the model has no equivalent of the counterfactual, causal step: it does not *reason* that gravity is the mechanism responsible and that any comparable object would behave identically for that reason, it merely reproduces the sequence of words most statistically associated with the prompt [4]. On his account, the fluency of a model's output can therefore mask the absence of anything resembling explanation or reasoning underneath it—a distinction directly relevant to trusting an LLM's generated code: producing a statistically plausible solution is not the same as having reasoned about why that solution is correct.

4 Can You Learn with AI tools?

Although GenAI tools do not encompass human intelligence, this does not mean they are not useful. A calculator does not possess intelligence, yet most people use one for calculations daily. However, we do not hand children calculators to teach them arithmetic—which raises the question of whether one can learn to program by using ChatGPT.

4.1 AI Reliance

To explore this notion of learning with GenAI, we look to a spectrum of AI reliance—to what extent do you use such tools and how does that impact your learning outcomes?

Blind Usage and No Usage

The two ends of this spectrum are *blind usage*—indiscriminately copy-pasting whatever the tool generates without critically evaluating its output—and *no usage*—simply not using any such tool at all. These two extremes are fairly easy to reason about in terms of learning outcomes. Someone who does not use AI tools learns exactly as students did before such tools became commercially available in 2022. Someone who uses them blindly, on the other hand, arguably learns next to nothing.

The mere possibility of blind usage is nonetheless worth noting. That it is possible to generate an entire solution to essentially anything asked in the CS1 course [**?**] is important to highlight, since any student under pressure from multiple looming deadlines might be pushed toward this end of the spectrum. Being aware, in advance, that this shortcut exists—and that taking it yields no learning—is something that can help a student avoid falling into it.

4.2 AI as a Helper

The more interesting part of this AI reliance spectrum, however, is what lies between the two extremes—using AI as an aid in the learning process. This could include using such tools to research a topic, ideation, have concepts explained, get help writing text, or better understand existing code. What learning outcomes result from this kind of usage?

Cognitive Ease at a Cost

Stadler et al. [13] investigated the cognitive load—the mental effort imposed on working memory during information processing—when students used GenAI tools compared to traditional search engines in learning tasks. Cognitive load is typically distinguished into intrinsic, extraneous, and germane load, where effective learning depends on managing these components appropriately [14]. While reducing unnecessary (extraneous) cognitive load is beneficial, sufficient engagement in germane cognitive processes—those directly related to constructing and refining knowledge—is essential for deep learning.

ChatGPT assistance was shown to reduce the effort associated with information retrieval and synthesis, thereby lowering overall cognitive load during task completion. However, this reduction appears to come at a cost. By streamlining access to ready-made explanations and solutions, GenAI tools may limit the extent to which learners actively engage in sense-making processes, such as evaluating sources, integrating information, and constructing their own understanding [14].

Your Brain on ChatGPT

Kosmyna et al.'s study *Your Brain on ChatGPT* [8] examined how access to AI tools influences writing processes and subsequent recall. 54 adult participants were asked to write essays within a set time frame across three sessions, each assigned throughout to one of three conditions:

- GenAI-assisted writing (using ChatGPT)
- Search engine-assisted writing (with access to Google)
- Unassisted writing (brain-only condition)

A notable finding from the study was the difficulty participants in the GenAI condition experienced in accurately recalling their own work. Immediately after finishing the first essay-writing session, participants were asked to quote a passage from what they had just written. At this first session, 83.3% of participants in the GenAI group were unable to produce an accurate quote from an essay they had completed only minutes earlier, compared to 11.1% in each of the other two groups [8]. Throughout all writing sessions, the Search Engine and Brain-only groups exhibited significantly fewer issues with quoting than those using GenAI.

5 How should the students approach CS1?

The final part of the lecture is an explanation of why I have just spent an hour talking about AI: students need to be informed of the pitfalls of these tools before starting this course, where an uncritical, over-reliant use of GenAI would quietly undermine the very skills the course exists to build. I then go on to outline the official guidelines given for all assignments and the exam of the course: *no use of GenAI is allowed*.

6 Conclusion - How did the Lecture go?

It has now been three weeks since the lecture was given...

References

[1] Artificial intelligence. Accessed: 2026-05-08.

[2] Deep blue. Accessed: 2026-03-08.

[3] Krzysztof Budzyń, Marcin Romańczyk, Diana Kitala, Paweł Kołodziej, Marek Bugajski, Hans O Adami, Johannes Blom, Marek Buszkiewicz, Natalie Halvorsen, Cesare Hassan, Tomasz Romańczyk, Øyvind Holme, Krzysztof Jarus, Shona Fielding, Melina Kunar, Maria Pellise, Nastazja Pilonis, Michał Filip Kamiński, Mette Kalager, Michael Bretthauer, and Yuichi Mori. Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study. *The Lancet Gastroenterology Hepatology*, 10(10):896–903, 2025.

[4] Noam Chomsky, Ian Roberts, and Jeffrey Watumull. Noam chomsky: The false promise of chatgpt. *The New York Times*, 8(4):177–179, 2023.

[5] Unesco. General Conference. *Revised recommendation concerning the international standardization of educational statistics adopted by the General Conference at its twentieth session, Paris, 27 November 1978.* Unesco, 1978.

[6] Stefan Feuerriegel, Jochen Hartmann, Christian Janiesch, and Patrick Zschech. Generative ai: S. feuerriegel et al. *Business & information systems engineering*, 66(1):111–126, 2024.

[7] James Finnie-Ansley, Paul Denny, Brett A. Becker, Andrew Luxton-Reilly, and James Prather. The robots are coming: Exploring the implications of openai codex on introductory programming. In *Proceedings of the 24th Australasian Computing Education Conference*, ACE '22, page 10–19, New York, NY, USA, 2022. Association for Computing Machinery.

[8] Nataliya Kosmyna, Eugene Hauptmann, Ye Tong Yuan, Jessica Situ, Xian-Hao Liao, Ashly Vivian Beresnitzky, Iris Braunstein, and Pattie Maes. Your brain on chatgpt: Accumulation of cognitive debt when using an ai assistant for essay writing task. *arXiv preprint arXiv:2506.08872*, 4, 2025.

[9] Duri Long and Brian Magerko. What is ai literacy? competencies and design considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, page 1–16, New York, NY, USA, 2020. Association for Computing Machinery.

[10] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *arXiv preprint arXiv:2203.02155*, 2022.

[11] Claude E. Shannon. Xxii. programming a computer for playing chess. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 41(314):256–275, 1950.

[12] Judy Hanwen Shen and Alex Tamkin. How ai impacts skill formation, 2026.

[13] Matthias Stadler, Maria Bannert, and Michael Sailer. Cognitive ease at a cost: Llms reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior*, 160:108386, 2024.

[14] John Sweller. Chapter two - cognitive load theory. volume 55 of *Psychology of Learning and Motivation*, pages 37–76. Academic Press, 2011.

[15] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.